

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Computer Science and Engineering) SEMESTER - 2 Summer 2025 (Regular)

Course :Bachelor of Technology (Computer Science and Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 2

Subject Code & Name: 24AF1CHEBS202 - ENGINEERING CHEMISTRY

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 Disinfection of water is done by...
a) Chlorine b) Bromine c) Iodine d) None
- 2 Temporary hardness in water can be removed by ...
a) Filtration b) Boling c) Sedimentation d) None
- 3 The exhausted cation exchange column is regenerated by ...
a) dil HCl b) dil KOH c) NaOH d) NaCl
- 4 A good fuel should possess high ...
a) Moisture b) Ash c) Sulphur d) Carbon
- 5 In Proximate Analysis of coal, the moisture content is determined at a temperature ...
a) 75°C b) 105°C c) 25°C d) 725°C
- 6 When graphite is dispersed in oil, it is called as.....
a) Grease b) Aquadag c) Oildag d) Blended oil
- 7 A fuel cell is used to convert chemical energy into energy
a) Mechanical b) Solar c) Electrical d) Potential
- 8 Phenolphthalein is.....
a) strong acid b) weak acid c) strong base d) weak base
- 9 Functional groups in a compound can be established by using
a) IR Spectroscopy b) UV Spectroscopy c) Chromatography d) None
- 10 Natural rubber is basically a polymer of
a) Isoprene b) Propylene c) Ethylene d) Propane
- 11 According to Beer's law, absorbance is equal to
a) $\log(I_o)$ b) $\log(I_o/I_t)$ c) $\epsilon.C.t$ d) Both b & c
- 12 Major component of Portland cement is.....
a) tricalcium silicate b) CaO c) MgO d) Calcium Sulphate

Q2. Solve the following.

- A) Explain Ion-Exchange process for softening of hard water with neat labeled diagram.

- B) Define the term Dissolved Oxygen and Explain the Winkler method to determine DO. 6
- Q3. Solve the following.
- A) What is fuel? Write its classification. Explain any four Characteristics of a good fuel. 6
- B) Define Calorific value. Write the principle of Boy's calorimeter and Calculate higher and lower calorific value per m^3 at STP from following data obtained in Boy's calorimeter Volume of gas used = 0.14 m^3 , Weight of water heated = 20 Kg, Temp. of Inlet Water = 23°C , Temp. of Inlet Water = 33°C , Mass of Steam Condensed = 0.020 Kg, Take heat liberated in condensing water vapour and cooling the condensate as 580 KCal/Kg 6
- Q4. Solve Any Two of the following.
- A) Write a note on: Glass electrode. 6
- B) What is an indicator? Explain Ostwald's theory of acid-base indicator. 6
- C) What is Conductometric titration? Write its principle. Explain conductometric titration of Strong acid Vs strong base with diagram. 6
- Q5. Solve Any Two of the following.
- A) What is Beer's law?. Derive an expression for Beer's law. 6
- B) What is chromatography? Explain thin layer chromatography in details & write its application. 6
- C) Explain the working of double beam IR Spectrophotometer with the schematic diagram. Write applications of IR Spectroscopy. 6
- Q6. Solve Any Two of the following.
- A) Define polymer. Explain thermoplastic and thermosetting resins. 6
- B) What is Cement? Write the Chemical Composition of Portland cement. Explain the functions of constituents in cement. 6
- C) Write synthesis of Urea formaldehyde and its properties & applications. 6

*** End ***